9 Vector Fields

Smoothness, Lie Brackets & Integral Curves

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Vector Fields

Smooth curves on a closed interval

Definition (Smooth curve)

A mapping $\sigma : [a, b] \to M$ is a *smooth curve* in M if σ extends to a C^∞-map of $(a - \varepsilon, b + \varepsilon) \to M$ for some $\varepsilon > 0$.

Note

If $\sigma : [a, b] \to M$ is a smooth curve, then its tangent vector

$$\dot{\sigma}(t) = \sigma_* \left(\frac{d}{dt}\Big|_t \right) \in T_{\sigma(t)} M$$

is well-defined for all $t \in [a, b]$.

Vector fields along a curve

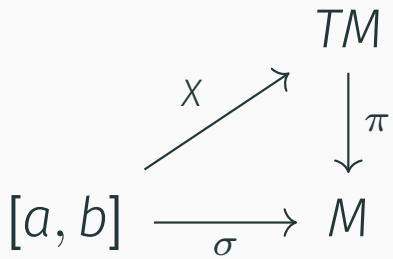
Definition (Vector field along a curve)

A smooth vector field X along a curve $\sigma : [a, b] \to M$ is a C^∞-map

$$X : [a, b] \longrightarrow TM$$

which *lifts* σ, i.e.

$$\pi \circ X = \sigma.$$



Vector fields on an open subset

Definition (Vector fields on an open subset of M)

A smooth vector field on an open set U in M is a C^∞-map $X : U \to TM$ s.t.

$$\pi \circ X = \mathrm{id}_U.$$

Sometimes we call a smooth vector field X on M a *section* of TM. The set of all smooth sections on U of TM is denoted by $\Gamma(U; TM)$. It is a vector space over $\mathbb{R}$ and also a $C^\infty(U)$-*module*:

$$\Gamma(U; TM) = C^\infty(U, TM) = \mathfrak{X}(U).$$

A Criterion for Smoothness

Three equivalent conditions

Proposition

Let X be a vector field on M. Then the following are equivalent:

(a) *X is C^∞;*

(b) *if $(U, x_1, \ldots, x_d)$ is a local coordinate system on M and $\{a_i\}$ are the functions on U defined by*

$$X|_U = \sum_{i=1}^{d} a_i \frac{\partial}{\partial x_i},$$

then $a_i \in C^\infty(U)$;

(c) *whenever $V \subset M$ is open, if $f \in C^\infty(V)$, then $X(f) \in C^\infty(V)$.*

Proof: (a) $\Rightarrow$ (b)

Let $X : M \to TM$ be a C^∞-map s.t. $\pi \circ X = \mathrm{id}_M$, so $X(p) = V_p \in T_pM$. Take a local coordinate system $(U, \varphi, x_1, \ldots, x_d)$ about $p \in M$; the local coordinate system about V_p is given by the local trivialization $\widetilde{\varphi} : \pi^{-1}(U) \to \mathbb{R}^{2d}$.

Smoothness of X in local coordinates requires $\widetilde{\varphi} \circ X \circ \varphi^{-1}$ to be C^∞ from $\varphi(U) \subset \mathbb{R}^d$ to $\widetilde{\varphi}\big(\pi^{-1}(U)\big) \subset \mathbb{R}^{2d}$. Writing $X(p) = V_p = \sum_i a_i(p) \left.\frac{\partial}{\partial x_i}\right|_p$,

$$\begin{aligned}\widetilde{\varphi} \circ X \circ \varphi^{-1}(x_1, \ldots, x_d) &= \Big(\varphi\big(\varphi^{-1}(x_1, \ldots, x_d)\big);\ a_i\big(\varphi^{-1}(x_1, \ldots, x_d)\big)\Big) \\ &= \big(x_1, \ldots, x_d;\ a_i(\varphi^{-1}(x_1, \ldots, x_d))\big),\end{aligned}$$

which means $a_i \in C^\infty(U)$.

Proof: (b) $\Rightarrow$ (c) $\Rightarrow$ (a)

(b) $\Rightarrow$ (c). For $f \in C^\infty(V)$ we show $X(f)$ is C^∞ about each $p \in V$. Take a local coordinate system $(U, x_1, \ldots, x_d)$ about p; then $X\big|_U = \sum_i a_i \frac{\partial}{\partial x_i}$ with $a_i \in C^\infty(U)$, so

$$X(f) = \sum_{i=1}^{d} a_i \frac{\partial f}{\partial x_i} = \sum_{i=1}^{d} a_i \frac{\partial (f \circ \varphi^{-1})}{\partial x_i} \in C^\infty(U).$$

(c) $\Rightarrow$ (a). In an arbitrary local coordinate system $(U, x_1, \ldots, x_d)$, write $X = \sum_i a_i \frac{\partial}{\partial x_i}$. Apply (c) to the coordinate function $x_i \in C^\infty(U)$:

$$X(x_i) = a_i \in C^\infty(U).$$

The Lie Bracket

The Lie bracket of two vector fields

If X, Y are smooth vector fields on M (i.e. $X, Y \in \mathfrak{X}(M) = \Gamma(TM)$), we define a vector field $[X, Y]$, the *Lie bracket* of X and Y, by

$$[X, Y]_p(f) = X_p\big(Y(f)\big) - Y_p\big(X(f)\big).$$

Note

The bracket measures the *failure* of the flows of X and Y to commute; locally it is the commutator of the derivations X and Y.

Properties of the Lie bracket

Proposition

Let $X, Y, Z \in \mathfrak{X}(M)$ and $f, g \in C^\infty(M)$.

(a) $[X, Y]$ *is a smooth vector field on* M;

(b) $[fX, gY] = fg\,[X, Y] + f\,(Xg)\,Y - g\,(Yf)\,X$;

(c) $[X, Y] = -[Y, X]$ *(skew-symmetry);*

(d) $\big[[X, Y], Z\big] + \big[[Y, Z], X\big] + \big[[Z, X], Y\big] = 0$ *(Jacobi identity).*

Note (Lie algebra)

A vector space with a bilinear operation satisfying (c) and (d) is called a *Lie algebra*.

Integral Curves

Integral curves

Definition (Integral curve)

Let $X \in \mathfrak{X}(M)$. A smooth curve σ in M is an *integral curve* of X if

$$\dot{\sigma}(t) = X\big(\sigma(t)\big) \qquad (\forall\, t \in I).$$

Reformulation

A curve $\gamma : (a, b) \to M$ is an integral curve of $X \iff$

$$\gamma_*\Big(\frac{d}{dt}\Big|_t\Big) = X\big(\gamma(t)\big) \qquad (t \in (a, b)).$$

Existence and uniqueness: local form

Suppose $0 \in (a, b)$ and $\gamma(0) = p$. Choose a local coordinate system (U, φ) with coordinate functions $x_1, \ldots, x_d$ about p; write

$$X|_U = \sum_{i=1}^{d} f_i \frac{\partial}{\partial x_i} \qquad (f_i \in C^\infty(U)).$$

For all t with $\gamma(t) \in U$,

$$\gamma_*\left(\frac{d}{dt}\Big|_t\right) = \sum_{i=1}^{d} \frac{d(x_i \circ \gamma)}{dt}\Big|_t \left.\frac{\partial}{\partial x_i}\right|_{\gamma(t)} = X(\gamma(t)) = \sum_{i=1}^{d} f_i(\gamma(t)) \left.\frac{\partial}{\partial x_i}\right|_{\gamma(t)}.$$

Reduction to a system of ODEs

Thus γ is an integral curve of X on $\gamma^{-1}(U)$ iff, with $\gamma_i \triangleq x_i \circ \gamma$,

$$(\star) \qquad \left.\frac{d\gamma_i}{dt}\right|_t = f_i\big(\varphi^{-1}\big(\gamma_1(t), \ldots, \gamma_d(t)\big)\big) \qquad (t \in \gamma^{-1}(U)).$$

Theorem (Existence & uniqueness)

($\star$) is a system of ODEs, for which there exists a fundamental existence & uniqueness theorem; hence integral curves of a smooth vector field exist and are unique.

Existence and Uniqueness

Integral curves: the question

Definition (Integral curve)

Let X be a smooth vector field on M. A smooth curve σ in M is an *integral curve* of X if

$$\dot{\sigma}(t) = X\big(\sigma(t)\big)$$

for each t in the domain of σ.

Question

Given a smooth vector field on M and a point $p \in M$, does there exist an integral curve of X through p? If so, is there a unique one?

The ODE system in local coordinates

We interpret in coordinates! Let $0 \in (a,b)$, $p = \sigma(0)$. Let $(U, \varphi = (x_1, \dots, x_d))$ be a local coordinate system (centered at p). Then the vector field X can be written as

$$X = \sum_{i=1}^{d} f_i \frac{\partial}{\partial x_i}, \qquad f_i \in C^\infty(U).$$

Assuming $\sigma\big((a,b)\big) \subset U$,

$$\dot\sigma(t) = \sigma_*\Big(\frac{d}{dt}\Big|_t\Big) = \sum_{i=1}^{d} \frac{d(x_i \circ \sigma)}{dt} \left.\frac{\partial}{\partial x_i}\right|_{\sigma(t)} \qquad \forall\, t \in (a,b).$$

So $\dot\sigma(t) = X\big(\sigma(t)\big)$ iff

$$\frac{d\sigma_i}{dt} = f_i\big(\varphi^{-1}\big(\sigma_1(t), \dots, \sigma_d(t)\big)\big), \quad i = 1, \dots, d. \tag{1}$$

Reduction to ODEs

Here $(\sigma_1, \ldots, \sigma_d) = (x_1 \circ \sigma, \ldots, x_d \circ \sigma) \in \varphi(U)$ are the coordinates of the point $\sigma(t) \in M$.

(1) is a system of ODEs, for which there exist fundamental existence and uniqueness theorems. In terms of the terminology for manifolds, we have:

Theorem

Let X be a C^∞ vector field on M. For each $p \in M$ there exist $a(p)$, $b(p) \in \mathbb{R} \cup \{\pm\infty\}$ and a smooth curve $\sigma_p : (a(p), b(p)) \to M$ such that

(a) $0 \in (a(p), b(p))$ *and* $\sigma_p(0) = p$;

(b) σ_p *is an integral curve of* X;

(c) *if* $\mu : (c, d) \to M$ *satisfies (a), (b), then* $(c, d) \subset (a(p), b(p))$ *and* $\sigma_p|_{(c,d)} = \mu$.

Proof of uniqueness and maximality

If α, β are integral curves defined on open intervals I, J, assume they have the same initial condition $\alpha(t_0) = \beta(t_0)$ for the same $t_0 \in I \cap J$. Then $I \cap J \neq \varnothing$. Define

$$K = \big\{ t \in I \cap J \;\big|\; \alpha(t) = \beta(t) \big\}.$$

- K is *open* (by the uniqueness theorem of ODE);
- K is *closed* (by continuity);
- hence $K = I \cap J$ (by the connectedness of $I \cap J$).

Thus (c) is true — (a), (b) as well: $\big(a(p), b(p)\big)$ is the union of all domains containing $t = 0$ on which an integral curve with $\sigma(0) = p$ exists, and it is non-empty by the fundamental existence theorem.

The Flow of a Vector Field

The transformations X_t

Definition (X_t and its domain)

$\forall\, t \in \mathbb{R}$, we define a transformation X_t with domain

$$\mathcal{D}_t = \left\{ p \in M \;\middle|\; t \in \big(a(p), b(p)\big) \right\},$$

by setting

$$X_t(p) = \sigma_p(t).$$

X_t moves each point p for time t along its maximal integral curve.

Properties of the flow

Theorem (Properties of the flow)

(d) $\forall\, p \in M$, $\exists$ *a neighborhood* V *of* p *and* $\varepsilon > 0$ *s.t. the map* $(t, p) \longmapsto X_t(p)$ *is defined and* C^∞ *from* $(-\varepsilon, \varepsilon) \times V \to M$;

(e) $\mathcal{D}_t$ *is open for each* t;

(f) $\bigcup_{t>0} \mathcal{D}_t = M$;

(g) $X_t : \mathcal{D}_t \longrightarrow \mathcal{D}_{-t}$ *is a diffeomorphism with inverse* X_{-t};

(h) *let* $s, t \in \mathbb{R}$. *Then the domain of* $X_s \circ X_t$ *is contained in but generally not equal to* $\mathcal{D}_{s+t}$. *However, the domain of* $X_s \circ X_t$ *is* $\mathcal{D}_{s+t}$ *in the case in which* s, t *have the same sign. And we have*

$$X_s \circ X_t = X_{s+t}.$$

Proof of the Flow Properties

Smooth dependence and (f)

(d) holds true according to the ODE theorem that the solution depends on both t and the initial value $\varphi(p)$ smoothly. So there's a neighborhood V of p and $(-\varepsilon, \varepsilon)$ of $t = 0$ s.t. $\mathcal{D}_t$ is defined for all $t \in (-\varepsilon, \varepsilon)$ and $V \subset \mathcal{D}_t$.

(f) Let $\big(a(p), b(p)\big)$ be the union of the domains of t that contain $t = 0$ and on which $\sigma(t)$ is an integral curve of X with $\sigma(0) = p$. By the fundamental existence theorem, $\big(a(p), b(p)\big) \neq \varnothing$. Therefore

$$\bigcup_{t>0} \mathcal{D}_t = M. \quad \square$$

Proof of (h): shifting the initial point

Let $t \in \big(a(p), b(p)\big)$. Then $s \longmapsto \sigma_p(t+s)$ is an integral curve of X initiated at $\sigma_p(t)$, with maximal domain $\big(a(p)-t,\, b(p)-t\big)$. By (c),

$$\big(a(p)-t,\, b(p)-t\big) = \big(a\big(\sigma_p(t)\big),\, b\big(\sigma_p(t)\big)\big).$$

So

$$\sigma_p(t+s) = \sigma_{\sigma_p(t)}(s). \tag{2}$$

Now, if $p \in$ the domain of $X_s \circ X_t$, then

$$t \in \big(a(p), b(p)\big) \quad \text{and} \quad s \in \big(a\big(\sigma_p(t)\big),\, b\big(\sigma_p(t)\big)\big) = \big(a(p)-t,\, b(p)-t\big).$$

So $s+t \in \big(a(p), b(p)\big)$, i.e. $p \in \mathcal{D}_{s+t}$, and we know $X_s \circ X_t = X_{s+t}$ by (2).

Opposite signs: the domain can shrink

For s and t that do not have the same sign, the inclusion may be strict. **Eg.** On $\mathbb{R}^2 \setminus \{0\}$, consider $\frac{\partial}{\partial r_1}$ and take $s = -1$, $t = 1$. Then the domain of

$$X_{-1} \circ X_1 \quad \text{is} \quad \mathbb{R} \setminus [-1, 0],$$

but

$$\mathcal{D}_{-1+1} = \mathcal{D}_0 = \mathbb{R} \setminus \{0\}.$$

Note

The composed map can be defined *less* often than X_{s+t}: a point may leave the manifold at time t and the way back at time s need not exist.

Same sign: the domain of $X_s \circ X_t$ is $\mathcal{D}_{s+t}$

If, however, s, t have the same sign, for eg. $s, t > 0$: $\forall\, p \in \mathcal{D}_{s+t}$,

$$s+t \in \big(a(p), b(p)\big) \;\Rightarrow\; \left.\begin{array}{l} t \in \big(a(p), b(p)\big) \\ s \in \big(a(p) - t,\, b(p) - t\big) = \big(a\big(\sigma_p(t)\big),\, b\big(\sigma_p(t)\big)\big) \end{array}\right\}$$

$$\Rightarrow\; p \in \mathcal{D}_t, \quad \sigma_p(t) \in \mathcal{D}_s \;\Rightarrow\; p \in \text{domain of } X_s \circ X_t.$$

(e) is trivial when $t = 0$, i.e. $\mathcal{D}_0 = M$ is open. Assume that $t > 0$ and $p \in \mathcal{D}_t$ ...

Completion of the Flow Proofs

Recall (h): shifting the initial point

Let $t, s \in \mathbb{R}$. For any p in the domain of $X_s \circ X_t$ we see that

$$p \in \mathcal{D}_t \quad \big(t \in \big(a(p), b(p)\big)\big) \quad \text{and} \quad \sigma_p(t) = X_t(p) \in \mathcal{D}_s.$$

To see that $p \in \mathcal{D}_{s+t}$, notice that we can define a C^∞-map $\gamma : s \longmapsto \sigma_p(t+s)$. Then

$$\gamma(0) = \sigma_p(t), \qquad \dot{\gamma}(s) = X\big(\sigma_p(t+s)\big) = X\big(\gamma(s)\big).$$

So $\big(a(\sigma_p(t)), b(\sigma_p(t))\big)$ is the maximal interval for γ, hence equals $\big(a(p) - t,\, b(p) - t\big)$, and

$$\sigma_p(t+s) = \sigma_{\sigma_p(t)}(s). \tag{2}$$

So if $p \in \mathcal{D}_t$ and $\sigma_p(t) \in \mathcal{D}_s$ then $p \in \mathcal{D}_{t+s}$. The inverse *doesn't* hold! For eg. the domain of $X_{-0.9} \circ X_1$ is contained in $\mathcal{D}_{0.1}$ but not equal to it.

Proof of (e): $\mathcal{D}_t$ is open — setup

Obviously true when $t = 0$, since $\mathcal{D}_0 = M$. Fix $t > 0$ (similar for $t < 0$) and $p \in \mathcal{D}_t$. We need a neighborhood N_p s.t. $N_p \subset \mathcal{D}_t$, i.e. $\forall\, q \in N_p$, $t \in \big(a(q), b(q)\big)$.

Consider the curve $\sigma_p\big([0,t]\big)$, which is *compact* in M, and let

$$\Gamma = \big\{ \big(\tau, \sigma_p(\tau)\big) \;\big|\; \tau \in [0,t] \big\}.$$

By (d), for each point $\big(t_0, \sigma_p(t_0)\big) \in \Gamma$ there is an $\varepsilon_{t_0} > 0$ and $V_{\sigma_p(t_0)}$ open about $\sigma_p(t_0)$ s.t. $(-\varepsilon_{t_0}, \varepsilon_{t_0}) \times V_{\sigma_p(t_0)}$ carries the flow.

Proof of (e): compactness and induction

By the compactness of Γ, there is a finite subcover for $\sigma_p([0,t])$: an interval $(-\varepsilon,\varepsilon)$ and a neighborhood W of $\sigma_p([0,t])$ s.t. the flow is defined on $(-\varepsilon,\varepsilon)\times W$. Let n be an integer >1 s.t. $\left|\frac{t}{n}\right|<\varepsilon$. Define

$$W_1=\left\{p\in W \;\middle|\; X_{\frac{t}{n}}(p)\in W\right\}=W\cap X_{\frac{t}{n}}^{-1}(W),$$

which is open in W; inductively,

$$W_k=W\cap X_{\frac{t}{n}}^{-1}(W_{k-1})\subset W \quad \text{open}, \qquad p\in W_n.$$

W_n consists of all points remaining in W after n translations, so $W_n\subset\mathcal{D}_t$. Therefore $\mathcal{D}_t$ is open.

Proof of (g): X_t is a diffeomorphism

$X_t : \mathcal{D}_t \longrightarrow \mathcal{D}_{-t}$ with inverse X_{-t}. It follows from

$$X_t = X_{\frac{t}{n}} \circ \cdots \circ X_{\frac{t}{n}},$$

with each $X_{\frac{t}{n}}$ C^∞ on a neighborhood of $p \in \mathcal{D}_t$, that X_t is C^∞.

X_t is 1:1 because

$$X_{-t} \circ X_t(p) = \sigma_{\sigma_p(t)}(-t) = p$$

by the uniqueness of the initial-value problem of ODE. And X_{-t} is also C^∞.

Complete Vector Fields

Complete vector fields and 1-parameter groups

Definition (Complete vector field)

A smooth vector field X on M is *complete* if $\mathcal{D}_t = M$ for all t (i.e. all the points of M can be translated up till time t).

- The set of transformations $\{X_t \mid t \in \mathbb{R}\}$ forms a *group* of transformations, isomorphic to $(\mathbb{R}, +)$, because $X_t \circ X_s = X_{t+s}$.
- If X is not complete, the transformations X_t do *not* form a group since the domain may depend on t. In this case we refer to the collection $\{X_t\}$ as the *local 1-parameter group* of X.

Remark: a simple noncomplete example

Consider $\frac{\partial}{\partial r_1}$ on $\mathbb{R}^2 \setminus \{0\}$. The integral curve σ passing through (a, b) is given by

$$\begin{cases} \dot{\sigma}(t) = \left.\dfrac{\partial}{\partial r_1}\right|_{\sigma(t)} \\ \sigma(0) = (a, b) \end{cases} \iff \begin{cases} \dfrac{dr_1}{dt} = 1 \\ \dfrac{dr_2}{dt} = 0 \end{cases} \quad (r_1, r_2)(0) = (a, b).$$

So the integral curve starting from $(a, 0)$ $(a > 0)$ has a life span $(-a, +\infty)$.

Note

If the manifold M is *compact*, any C^∞ vector field on M is complete! (Exercise !!)

Straightening a Vector Field

Prop 1.53: straightening out a nonzero vector field

Proposition 1.53

Let $p \in M^d$, and let X be a smooth vector field on M s.t. $X(p) \neq 0$. Then there is a coordinate system $(U, \varphi) = (U, x_1, \ldots, x_d)$ about p s.t.

$$X|_U = \left.\frac{\partial}{\partial x_1}\right|_U.$$

Near a non-vanishing point, X can be taken as the first coordinate vector field.

Proof: the flow map

Choose a coordinate system centered at p, denoted by $(V, \psi) = (V, y_1, \ldots, y_d)$, s.t.

$$X|_p = \left.\frac{\partial}{\partial y_1}\right|_p.$$

Consider the flow defined by

$$\begin{cases} \dot{\sigma}(t) = X\big(\sigma(t)\big), \\ \sigma(0) = \psi^{-1}(0, \eta_2, \ldots, \eta_d). \end{cases}$$

Then $\exists\, \varepsilon > 0$ and a neighborhood $W \subset \mathbb{R}^{d-1}$ about 0 s.t.

$$\sigma(t, \eta_2, \ldots, \eta_d) = X_t\big(\psi^{-1}(0, \eta_2, \ldots, \eta_d)\big)$$

is defined and smooth for $(t, \eta_2, \ldots, \eta_d) \in (-\varepsilon, \varepsilon) \times W \subset \mathbb{R}^d$.

Proof: σ is non-singular at the origin

$$\sigma_{*,0}\Big(\frac{d}{dt}\Big|_0\Big) = X(p) = \frac{\partial}{\partial y_1}\Big|_p .$$

For $k = 1, \ldots, d$ and $i = 2, \ldots, d$,

$$\begin{aligned}
\sigma_{*,0}\Big(\frac{\partial}{\partial \eta_i}\Big|_0\Big)(y_k) &= \frac{d}{ds}\Big|_{s=0} y_k \circ \sigma(0, \ldots, s, \ldots, 0) \\
&= \frac{d}{ds}\Big|_{s=0} y_k \circ X_0\big(\psi^{-1}(0, \ldots, s, \ldots, 0)\big) \\
&= \frac{d}{ds}\Big|_{s=0} y_k \circ \psi^{-1}(0, \ldots, s, \ldots, 0) = \delta_{ki}.
\end{aligned}$$

So $\sigma_{*,0}\big(\frac{\partial}{\partial \eta_i}\big|_0\big) = \frac{\partial}{\partial y_i}\big|_p$, hence $\sigma_{*,0}$ is non-singular at p.

Proof: the new coordinates

By the inverse function theorem, $\exists\,(U, \sigma^{-1})$ about p. Denote $\sigma^{-1} = (x_1, \ldots, x_d)$. Then $\forall\, h \in C^\infty(U)$,

$$\begin{aligned}\left.\frac{\partial}{\partial x_1}\right|_U (h) &= \left.\frac{\partial}{\partial r_1}\right|_0 (h \circ \sigma) = \frac{\partial}{\partial t}(h \circ \sigma) \\ &= \sigma_*\left(\frac{\partial}{\partial t}\right)(h) = X(h) = X\big|_U(h).\end{aligned}$$